EXPERT OPINION ON DRUG METABOLISM & TOXICOLOGY
2021, VOL. 17, NO. 10, 1223–1235
https://doi.org/10.1080/17425255.2021.1982892

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REVIEW

The impact of CGRPergic monoclonal antibodies on prophylactic antimigraine therapy and potential adverse events

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ABSTRACT

Migraine is a prevalent medical condition and the second most disabling neurological disorder. Regarding its pathophysiology, calcitonin gene-related peptide (CGRP) plays a key role, and, consequently, specific antimigraine pharmacotherapy has been designed to target this system. Hence, apart from the gepants, the recently developed monoclonal antibodies (mAbs) are a novel approach to treat this disorder. In this review we consider the current knowledge on the mechanisms of action, specificity, safety, and efficacy of the above mAbs as prophylactic antimigraine agents, and examine the possible adverse events that these agents may trigger. Antimigraine mAbs act as direct scavengers of CGRP (galcanezumab, fremanezumab, and eptinezumab) or against the CGRP receptor (erenumab). Due to their long half-lives, these molecules have revolutionized the prophylactic treatment of this neurovascular disorder. Moreover, because of their physicochemical properties, these agents are hepato-friendly and do not cross the blood-brain barrier (highlighting the relevance of peripheral mechanisms in migraine). Nevertheless, apart from potential cardiovascular side effects, the interaction with AMY_1 receptors and immunogenicity induced by autoantibodies against mAbs could be a concern for the safety of long-term treatment with these molecules.

ARTICLE HISTORY
Received 29 April 2021
Accepted 15 September 2021

KEYWORDS
CGRP; eptinezumab; erenumab; fremanezumab; galcanezumab; migraine; monoclonal antibodies; pain; trigeminovascular system

1. Introduction

Migraine is considered a neurovascular disorder, and it basically involves a throbbing unilateral headache associated with other symptoms, including diarrhea, nausea, vomiting, pallor, blurred vision, phonophobia, photophobia, etc. (for refs. [1–4]). Although migraine can be classified as episodic or chronic (headache on ≥ 15 days per month for > 3 months) [2], this disorder can also be diagnosed as: (1) migraine with aura, which starts with sensory and/or motor symptoms (*i.e.* aura, followed by headache); and (2) migraine without aura [2,4].

From another perspective, in addition to representing a huge financial and/or psychological burden for the patient, family members, and society [5], migraine is the second leading cause of years of healthy life lost due to disability. Indeed, around the world, this disorder represents a total of > 1 billion people living with migraine, and the global age-standardized prevalence is higher in women (18.9%) than in men (9.8%) [6].

2. The pathophysiology and pharmacology of migraine: the role of 5-HT and CGRP

The history of the etiology and treatment of headaches (including migraine) is fascinating and extends over millennia [7]. This makes it an ancestral ailment that, little by little (*i.e.* from the Neolithic Age to the present day), has been identified as a neurovascular disorder [7]. Most importantly, in recent decades, great strides have been made in understanding the pathophysiology of migraine and in the development of specific antimigraine drugs [3,4,8–11]. As detailed and discussed in depth in other reviews [3,4,7–12], migraine pathophysiology is considered a complex neurovascular disorder that may involve at least two critical events in plasma, namely: (i) low levels of circulating serotonin (5-hydroxytryptamine; 5-HT); and/or (ii) high levels of circulating CGRP. These seminal findings, over the years, have paved the way for the successful development of more effective antimigraine compounds [3,4,7–12], such as: (i) triptans ($5\text{-HT}_{1B/1D}$ receptor agonists with pEC_{50} values [agonist activity] between 6.50 and 8.50 for 5-HT_{1F} receptors, such as sumatriptan and second-generation triptans) [8]; (ii) ditans (agonists at 5-HT_{1F} receptors, such as lasmiditan) [8]; (iii) gepants (potent and selective antagonists at CGRP receptors; e.g. rimegepant) [4,9]; and (iv) monoclonal antibodies [mAbs] against CGRP (*e.g.* fremanezumab) or its receptor (*e.g.* erenumab) [10–12]. Within this context, current pharmacotherapy against migraine can be broadly divided into specific treatments (*e.g.* ergots, triptans, ditans, gepants, and mAbs against CGRPergic transmission) and nonspecific treatments (*e.g.* non-steroidal anti-inflammatory drugs [NSAIDs], antiemetics, β-blockers, antiepileptics, etc.). In both cases, these treatments can be acute (triptans,

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Article highlights

- Globally, migraine is one of the most prevalent medical conditions and the second most disabling neurological disorder. Its pathophysiology has been strongly linked with a dysfunction of the CGRPergic neurotransmission in the trigeminovascular system.
- Consequently, drugs that specifically target CGRPergic neurotransmission are considered potential antimigraine drugs.
- Current specific antimigraine pharmacotherapies can be divided into: (i) acute abortive drugs, *e.g.* ergots, triptans, gepants, and ditans; and (ii) prophylactic drugs, *e.g.* CGRPergic monoclonal antibodies (mAbs).
- The triptans (5-HT$_{1B/1D}$ receptor agonists with pEC$_{50}$ values between 6.50 and 8.50 for 5-HT$_{1F}$ receptors, such as sumatriptan and second-generation triptans) and ditans (5-HT$_{1F}$ receptor agonists, such as lasmiditan) indirectly inhibit the release of CGRP from the trigeminovascular system, whereas the gepants (CGRP receptor antagonists) and mAbs were designed to directly target the CGRPergic transmission.
- In the case of mAbs, these agents act as direct scavengers of CGRP (galcanezumab, fremanezumab, and eptinezumab) or against the CGRP receptor (erenumab). Moreover, according to current randomized clinical trials (RCT's) conducted in patients with episodic and chronic migraine, these mAbs possess a good safety and efficacy profile.
- Indeed, considering the chemical nature of mAbs, these molecules are devoid of liver toxicity and seem to exert their actions outside the central nervous system (CNS).
- In any case, since CGRP may play a role in modulating the cardiovascular and other systems (*e.g.* gastrointestinal and cutaneous), there may still be some concerns about the use of anti-CGRP molecules with a long half-life.

This box summarizes key points contained in the article.

ditans, NSAIDs, etc.) or prophylactic (mAbs, β-blockers, antiepileptics, etc.) [9–12].

Undoubtedly, the triptans have been a breakthrough in migraine pharmacotherapy, but they are ineffective in some migraineurs, while in others, the cardiovascular side effects of triptans (even when effective) limit their use [3,8,10]. These problems led to the development of agonists at 5-HT$_{1F}$ receptors (ditans), including lasmiditan, a lipid-soluble molecule without vasoconstrictor properties (unlike the triptans) [8,10]. In addition, a new second generation of gepants (potent and selective CGRP receptor antagonists, *i.e.* rimegepant and ubrogepant) and novel mAbs against CGRP (*e.g.* fremanezumab) or its receptor (*e.g.* erenumab) have recently entered the arena of migraine pharmacotherapy.

Admittedly, new treatments have been emerging to treat migraine, but the current gold standard is represented by the triptans. These tryptamine derivatives, as previously reviewed [10,12], have a better therapeutic gain (pain freedom and pain relief at 2 h) in comparison with the gepants (*i.e.* rimegepant and ubrogepant) and lasmiditan. In contrast, the gepants seem to produce fewer side effects than the triptans, but lasmiditan produces several adverse events in the central nervous system (CNS) (*e.g.* dizziness, nausea, drowsiness, fatigue, and sensation of heaviness) that could limit its use. At this point, it is noteworthy that the first generation of gepants, including olcegepant, telcagepant, BI 44370 TA, and MK-3207, displayed some pharmacokinetic issues (bioavailability and/or hepatotoxicity) that halted their pharmaceutical development [4,9–11]. These drawbacks prompted the search for new strategies to directly target the CGRPergic system; hence CGRPergic mAbs emerged as an exciting approach. Certainly, given their physicochemical properties and very high molecular weight, they are not hepatotoxic and cannot cross the blood-brain barrier (BBB) [9–11]; so, they lack central side effects (see below).

Based on the above findings, this review will: (i) consider the current knowledge on the mechanisms of action, specificity, safety, and efficacy of the newly developed mAbs as prophylactic antimigraine agents; and (ii) analyze the potential adverse events that might be triggered by these agents when interfering with the CGRPergic system.

3. The rationale for the development of monoclonal antibodies (mAbs) as antimigraine agents

3.1. A brief historical overview

One of the main neuromodulators involved in the pathophysiology of migraine is CGRP. Indeed, seminal studies have shown that an intravenous infusion of this neuropeptide elicits vasodilatation and precipitates migraine attacks in subjects diagnosed with migraine [4]. Furthermore, during a migraine attack, an increase in plasma CGRP concentrations can be measured in the craniovascular circulation. In contrast, the triptans (the gold standard in migraine treatment) normalize these plasma CGRP levels, probably by prejunctional activation (at perivascular level) of 5-HT$_{1B/1D/1F}$ receptors [13,14]. Besides, it is well-known that, at the trigeminocervical complex (TCC), CGRP plays a crucial role in central sensitization (a critical step in the development of migraine headache), while the triptans inhibit CGRPergic transmission at this level [13]. This neuropeptide's vasodilator or pronociceptive effects are mediated by activating the CGRP receptor, a seven-transmembrane receptor coupled to G$_s$ proteins [15].

These data imply that the direct blockade of the CGRPergic transmission can be used as a pharmacological strategy to treat migraine. Under this assumption, clinical proof-of-efficacy for acute antimigraine treatment was shown with the nonpeptide CGRP receptor antagonists (*i.e.* the gepants) [16,17,18,19], which fostered the interest in this pharmacological target. Certainly, several clinical trials showed that first-generation gepants (*e.g.* telcagepant, BI 44370 TA, and MK-3207) were effective as an acute treatment for episodic migraine; notwithstanding, hepatotoxicity issues stopped their further pharmaceutical development (for refs. see [9]). Furthermore, second-generation gepants without hepatotoxicity effects have recently emerged and are currently used as acute antimigraine treatments (*i.e.* ubrogepant [Ubrelvy®] and rimegepant [Nurtec®], approved by the FDA in 2019 and 2020, respectively). Interestingly, another gepant without liver toxicity effects, namely, atogepant [18,19], has been tested in a Phase IIb/III randomized clinical trial (RCT), which showed that it could be used as a prophylactic antimigraine drug. Together, these data support the notion that molecules against the CGRPergic transmission are useful in migraine treatment [20,21].

On this basis, the role of CGRP in migraine pathophysiology and the chronic nature of this neurovascular disorder led to the idea that CGRPergic mAbs (proteins with high molecular weight, >100 kDa) could be used as a therapeutic strategy. Actually, as previously discussed by Bigal *et al*. [22], mAbs have several advantages over the gepants, namely: i) high specificity (*i.e*. less adverse events); ii) long terminal half-lives (*i.e*. less frequent dosing); and iii) lack of liver toxicity due to their metabolism.

Without a doubt, one of the first lines of evidence showing the potential use of mAbs against migraine was reported in 2007 by Juhl *et al*. [23]. In this ground-breaking work, which used a migraine rat model (*i.e*. intravital microscopy), the authors showed that an intravenous infusion of a humanized mAb against CGRP significantly inhibited the CGRP-induced vasodilatation in the dural artery; besides, this humanized mAb did not affect heart rate or arterial blood pressure in rats [20]. Together, these data were critical to support the hypothesis about using mAbs as antimigraine molecules. To date, the use of mAbs represents an innovative antimigraine treatment.

From a pharmacological perspective, the current mAbs against migraine target some part of the system associated with activation of G-protein coupled receptors (GPCRs). In the case of erenumab, this mAb binds and blocks the CGRP receptor, whereas galcanezumab, fremanezumab, and eptinezumab act as direct scavengers of peripheral CGRP [21]. In fact, erenumab represents the first class of large mAb molecules targeting GPCRs, which are the commonly exploited therapeutic targets by the traditional use of small molecules (<1 kDa).

3.2. A primer on the biology of mAbs as therapeutic agents

mAbs are immunoglobulins that can be used as therapeutic biotechnological agents for the treatment of various pathologies (*e.g*. autoimmune, neurodegenerative, viral). The hybridoma technology to produce specific mAbs against a particular antigen was rocketed by Kohler and Milstein's report [24] and quickly migrated from its use in basic molecular immunology research to its use as a diagnostic and therapeutic tool for cancer and autoimmune diseases [25]. Indeed, the first mAb approved as a therapeutic molecule was muromonab-CD3 in 1986 [26], which reduces the incidence of allograft rejection by binding to the T-cell receptor-CD3 complex [27]. Since that year, at least one hundred and six mAbs have been approved by the US Food and Drug Administration (FDA) and/or the European Medicines Agency (EMA). Furthermore, since 2018, from twenty-two mAbs approved by the FDA and EMA, four of these immunoglobulins (humanized and fully human mAbs) were authorized as migraine preventive treatments [28].

From a biological perspective, immunoglobulins (or antibodies) are specialized Y-shaped proteins produced by the immune system in response to antigens' intrusion. These molecules are composed of four protein chains (two identical light chains and two identical heavy chains) with a high molecular weight (~150 kDa) (Figure 1). In general, and based on their heavy chain sequences, immunoglobulins are broadly classified into five isotypes, namely, IgA, IgD, IgE, IgG, and IgM [29]. During an infection, the immune system produces a plethora of antibodies against several antigens' epitopes (a polyclonal antibody response due to activation of several B lymphocytes). In contrast, the hybridoma technique described in 1975 allows us to produce, from a single clone of B cells, specific mAbs to bind monospecifically to the same antigen epitope [30,31,32,33,34,35,36].

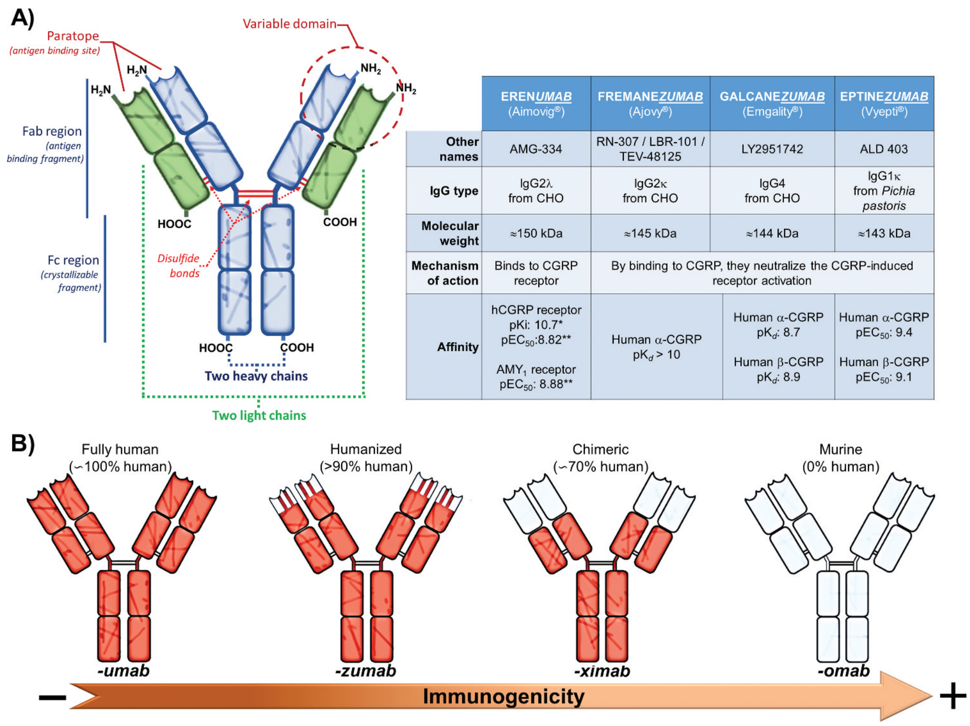
The main isotype used in therapeutics is IgG (γ-immunoglobulins) which, in humans, is subdivided into four subtypes, namely, IgG1, IgG2, IgG3, and IgG4, sharing nearly 90% of amino acids sequence [37]. At this point, one of the main issues in using mAbs is their capacity to induce an immune response; indeed, all mAbs can induce an immune response, mainly if the immunoglobulin targets some part of the immune system [38]. For example, muromonab-CD3 was synthesized from a murine hybridoma, and, consequently, after administration of this mAb, anti-murine antibodies are produced, limiting its use [39]. To decrease immunogenicity, the reduction of murine aminoacids sequences in mAbs fostered the advent of chimeric, humanized, and human mAbs [39–42]. Accordingly, for the International Nonproprietary Name (INN) system, mAbs can be murine (100% murine protein; *suffix-**omab***), chimeric (30% protein murine; *suffix-**ximab***), humanized (5–10% murine protein; *suffix-**zumab***), and fully human (~100% human protein; *suffix-**umab***). In all cases, the stem ***mab*** indicates a monoclonal antibody [43] (Figure 1).

Current therapeutic mAbs possess low immunogenicity (particularly for human and humanized mAbs) while exhibiting a high affinity for the target and large half-life values [44]. Moreover, it is to be highlighted that current mAbs against the CGRPergic transmission are classified as humanized or human mAbs, and were engineered to interact with CGRP or the CGRP receptor. Consequently, the probability of inducing a severe adverse immunological response seems very low [45].

3.3. Current CGRPergic mAbs approved for migraine treatment

Currently, four mAbs have been approved for migraine treatment [10]. These mAbs were designed for interrupting the CGRPergic transmission by targeting the CGRP receptor (*i.e*. erenumab) or by directly sequestering the circulating molecules of CGRP (*i.e*. galcanezumab, fremanezumab, and eptinezumab) [46,47] (see Figure 1 and Table 1 for details).

Erenumab (Aimovig®) is a fully human mAb that antagonized the CGRP receptor at 70 or 140 mg via a monthly subcutaneous injection [48]. It has been shown to increase the quality of life of patients with chronic migraine [49], and it also reduces the monthly migraine days (MMD) in women with menstrual migraine [50] and patients with episodic migraine [51]. It is essential to highlight that, in a 3-years cutoff from a 5-year open-label clinical trial, erenumab



	ERENUMAB (Aimovig®)	FREMANEZUMAB (Ajovy®)	GALCANEZUMAB (Emgality®)	EPTINEZUMAB (Vyepti®)
Other names	AMG-334	RN-307 / LBR-101 / TEV-48125	LY2951742	ALD 403
IgG type	IgG2λ from CHO	IgG2κ from CHO	IgG4 from CHO	IgG1κ from *Pichia pastoris*
Molecular weight	≈150 kDa	≈145 kDa	≈144 kDa	≈143 kDa
Mechanism of action	Binds to CGRP receptor	By binding to CGRP, they neutralize the CGRP-induced receptor activation		
Affinity	hCGRP receptor pKi: 10.7* pEC_{50}:8.82** AMY$_1$ receptor pEC_{50}: 8.88**	Human α-CGRP pK_d > 10	Human α-CGRP pK_d: 8.7 Human β-CGRP pK_d: 8.9	Human α-CGRP pEC_{50}: 9.4 Human β-CGRP pEC_{50}: 9.1

Figure 1. The biology of monoclonal antibodies (mAbs) and current FDA-approved mAbs against migraine. Panel (A) shows a general structure of the immunoglobulins or antibodies (Ab). Panel (B) illustrates the current nomenclature (i.e. – umab, – zumab, – ximab, and – omab) of monoclonal antibodies (mAbs) to define if a mAb is fully human, humanized, chimeric or murine; note that murine mAbs induce a strong immunogenicity response in comparison with fully human mAbs. In addition, the table shows the main characteristics of the current mAbs approved for migraine treatment. The data of this figure were taken from [9,12,32-36]. * Data from [33] suggest that erenumab does not have a relevant affinity for other calcitonin receptors, including AMY1 receptors (experiments performed in SK-N-M cells). In contrast, ** data from [91], showed that erenumab displays a similar potency activating human CGRP receptors (SK-N-M cells) or AMY1 receptors (HEK cells).

Table 1. An overview on the main pharmacokinetics and clinical characteristics of the current CGRPergic mAbs given monthly (m) or quarterly (q). For details see [9,11,48,55,59,62,63,66,105-108,116,150-155]. FDA, Food and Drug Administration; EMA, European Medicines Agency.

	Erenumab (Aimovig®)	Fremanezumab (Ajovy®)	Galcanezumab (Emgality®)	Eptinezumab (Vyepti®)
FDA /EMA date approval	May 2018 / Jul 2018	Sep 2018 / Mar 2019	Sep 2018 / Nov 2018	Feb 2020 / In Review
Approved for	Preventive treatment of migraine (episodic or chronic)			
Doses / route	70 mg (m) / s.c. 140 mg (m) / s.c.	225 mg (m) / s.c. 675 mg (q) / s.c.	120 mg (m) / s.c. 300 mg (m) / s.c.	100 mg (q) / i.v. 300 mg (q) / i.v.
Pharmacokinetics*	*Data from single 70 mg s.c. administration* C_{max}: 6.25 µg/ml T_{max}: 6 days $t_{1/2}$: 28 days AUC_{last}: 171 day Vd: 3.86 l (for 140 mg)	*Data from single 225 mg s.c. administration* T_{max}: ≈5 days $t_{1/2}$: 31 days AUC_{last}: 168 day Vd: 6 l	*Data from single 120 mg s.c. administration* T_{max}: 5 days $t_{1/2}$: ≈27 days Vd: 7 l	$t_{1/2}$: ≈27 days Vd: 3.7 l
Mean MMD before (*B*) and after (*A*) treatment or placebo (*P*)	*P* / *70 mg* / *140 mg*	*P* / *225 mg* / *675 mg*	*P* / *120 mg* / *240 mg*	*P* / *100 mg* / *300 mg*
	B 8.2 / 8.3 / 8.3	B 9.1 / 8.9 / 9.2	B 9.1 / 9.2 / 9.1	B 8.4 / 8.7 / 8.6
	A 6.4 / 5.1 / 4.6	A 6.5 / 4.9 / 5.3	A 6.3 / 4.5 / 4.5	A 5.2 / 4.8 / 4.3
Most common adverse events**	Injection site reactions (pain, erythema, pruritus) constipation, cramps/muscle spasm	Injection site reactions (pain, erythema, pruritus)	Injection site reactions (pain, erythema, pruritus), nasopharygitis	Upper respiratory tract infection, hypersensitivity
% Patients generating auto-antibodies***	4 – 8	0.3 – 0.7	3 - 5	< 8

* In general, the pharmacokinetics parameters were not influenced by age, sex, race, body weight or migraine subtype (episodic or chronic).
** Incidence of adverse events at least 2% greater than placebo (observed in RCTs).
*** Neutralizing and/or non-neutralizing.

showed good tolerability and a cardiovascular safety profile [52]. Erenumab (as many mAbs) does not induce, inhibit, or interact with CYP450 enzymes and has been proven safe against pharmacokinetics interactions with oral contraceptive drugs [48,53]. Another interesting feature of this mAb was its relatively persistent therapeutic effect that seems to last at least three months after treatment termination [54]. Nowadays, the price of Aimovig® is around USD 600 (https://www.aimovig.com/paying-for-aimovig). This cost is similar to that of the other approved mAbs against migraine (*e.g.* galcanezumab, whose price is around USD 600; https://www.lillypricinginfo.com/emgality). It is fair to point out that the developing countries' general population cannot afford the cost of these novel mAbs. Hence, accessibility, price, and distribution remain the main obstacle to receiving global developments in these technologies.

Galcanezumab (Emgality®) is a humanized mAb that binds and neutralizes the human CGRP molecule [55]. Galcanezumab is administered by subcutaneous injection in a single-dose prefilled pen or syringe with 120 mg and requires a loading dose of 240 mg with single monthly administrations of 120 mg [55]. It is well tolerated (with injection-site reactions being the major adverse event) and provides around a 30% reduction in the MMD of patients with episodic or chronic attacks [56]. Other studies have shown a similar decrease in the MMD in low – and high-frequency migraine headaches [57]. Galcanezumab is not metabolized by CYP450 enzymes [55], and it seems to have similar pharmacokinetics in the young and old population [58].

Fremanezumab (Ajovy®) is a fully-humanized immunoglobulin that targets CGRP molecules [59]. It provides a reduction in MMD [60] similar to that of galcanezumab. A dose of 225 mg monthly or 675 mg every three months via subcutaneous injection is recommended [59]. Fremanezumab follows the enzymatic proteolytic metabolism typical of the mAbs and does not involve CYP450 enzymes; thus, it is hypothetically safe against pharmacodynamic drug/nutrients interactions [59].

Lastly, **eptinezumab** (Vyepti®) is a humanized immunoglobulin that, via its binding with CGRP molecules, prevents the interaction of CGRP with its receptor [62]. The recommended dosage of eptinezumab in adults is 100–300 mg every three months. Metabolism of eptinezumab does not require CYP450 enzymes [62], decreasing the potential drug-drug and drug-nutrient interactions.

While migraine patients under preventive treatment still have to consume acute antimigraine drugs (*i.e.* ergots, triptans, and others) when they have an attack, eptinezumab can reduce the migraine risk attack up to 50% from baseline after an initial dose [63,64]; this treatment protected during 24 weeks with a second injection in week 12 with no severe side effects or significant complications reported [47,65]. Less than 8% of the participants who received eptinezumab 30–300 mg for up to 56 months in the PROMISE-1 study (*i.e.* 666 patients) developed neutralizing anti-eptinezumab antibodies [66].

3.4. What we have learned about the use of mAbs in other pathologies

Beyond any doubt, mAbs: (i) represent a revolutionary technology that has displaced the classical drugs as a more profitable investment for pharmaceutical companies; (ii) have become the best-selling drugs in the current world [67,68]; and (iii) represent >20% of the novel approved medications in the last years [46]. Over the economic aspects of this technology, mAbs represent hope for many people, as they focus on diseases that have not been treated for decades (*e.g.* HIV, Parkinson's disease, etc.) or even emergent infectious diseases such as COVID-19 [69–71].

One hypothetic advantage of mAbs over the classical antimigraine drugs is their high selectivity and, accordingly, lesser off-target adverse actions. Nevertheless, this technology does not lack some inconveniences related to the: (i) immunogenicity (the host immune response); (ii) immunotoxicity (adverse effects on the functioning of the immune system); and (iii) pharmacokinetics [72–74]. For example, immunogenicity, which is the consequence of introducing any xenobiotic to the organism (*i.e.* the mAbs themselves), was reduced after substituting the murine components for chimeric, humanized, or full human ones [75]. The consequences of immunogenicity are a rapid clearance that can prevent therapeutic effects and anaphylactic reactions in the highest risk scenarios [76]. Consequently, every mAbs-based drug should be carefully evaluated to avoid any adverse event related to chronic use [77,78].

4. The therapeutic impact of mAbs in the treatment of migraine

4.1. The site and mechanism of action of mAbs in migraine treatment

The CGRP receptor is widely expressed at the central and peripheral levels [79]. The activation of this receptor, coupled to G_s proteins, induces an increase of 3′,5′-cyclic adenosine monophosphate (cAMP), leading to peripheral vasodilatation and facilitation of nociceptive transmission in the trigeminocervical complex [80]. In general, all specific antimigraine drugs inhibit the CGRPergic transmission in the trigeminovascular system; this inhibition can be elicited indirectly (by ergots, triptans, and ditans) or directly (by gepants and anti-CGRPergic mAbs) [12].

Ergots, triptans, and ditans seem to exert their therapeutic effects by peripheral and central actions [81,82]; nevertheless, current evidence suggests that central actions are not required for producing their antimigraine effect. In favor of this view, a previous report using positron-emission tomography (PET) showed that telcagepant (a first-generation gepant that was *discontinued due to hepatotoxicity concerns*) produced no relevant receptor occupancy in the CNS [83]. Similar results were observed using ^{11}C-dihydroergotamine in a study using magnetic resonance imaging (MRI) [84]. Together, these data

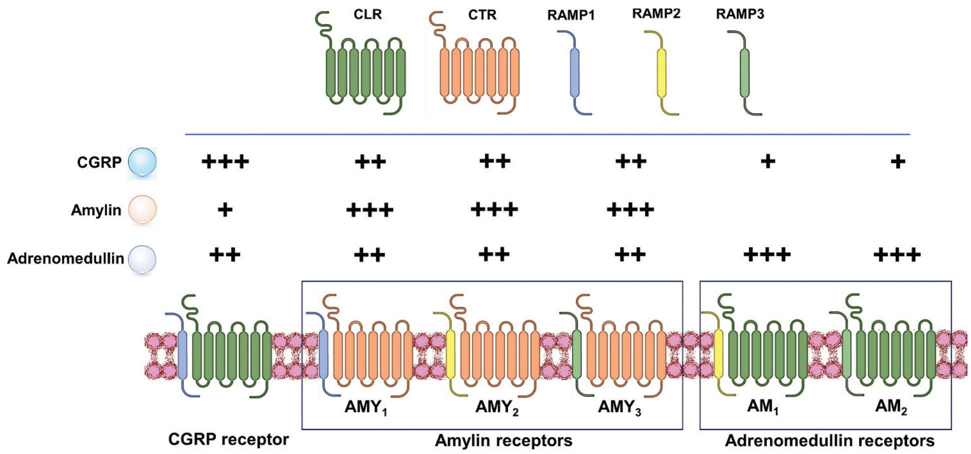


Figure 2. The calcitonin receptors (all coupled to Gs proteins) are atypical GPCR since, apart from the 7 transmembrane domains proteins (CLR or CTR), they require an accessory protein to be functional, i.e. the RAMPs. As illustrated, the different calcitonin receptors are responsive to CGRP, amylin and/or adrenomedullin. Abbreviations: GPCR, G-protein coupled receptors; CLR, calcitonin-like receptor; CTR, calcitonin receptor; RAMP, receptor activity-modifying protein.

imply that the CNS actions are not required to induce an antimigraine effect. This idea is clearly illustrated (and reinforced) in the case of mAbs.

Since mAbs are high molecular weight molecules (~150 kDa), their passage through the BBB is limited. Still, these molecules could interact with sites of the CNS uncovered by the BBB, like the trigeminal ganglion and paraventricular structures within the brainstem [85,86]. However, since only a low concentration of mAb in the CNS could be achieved after i.v. administration, central effects are probably minor, if any [87,88]; in fact, as recently reviewed, the CNS adverse events caused by current antimigraine mAbs seem to be inexistent [12]. In conjunction with the study of Hostetler *et al*. [83] and Schankin *et al*. [84], the consensus on the locus of action of these molecules is the peripheral part of the trigeminovascular system (*i.e*. at trigeminal ganglion and the intra – and extracranial blood vessels) [10].

Regarding the mAbs' mechanisms of action, although the – *numabs* and the – *zumabs* interfere with the CGRPergic system, erenumab was designed to target the CGRP receptor, whereas the – *zumabs* are scavengers of the CGRP molecules [89,90]. These minutiae about where mAbs act have a tremendous impact on how intracellular events (signaling and trafficking) might be elicited by targeting either the receptor or the peptide. For example, it is well-known that the interaction of CGRP with its receptor induces CGRP receptor internalization [61], and a recent report showed that erenumab (but not fremanezumab) induced CGRP receptor internalization [91]. At this point, we do not know the impact of the above differential effect on the CGRPergic system, but these data shed some light on the intricate pharmacological mechanisms that mAbs and gepants can induce.

As shown in Figure 2, the CGRP receptor, which belongs to the class B 'secretin-like' family of GPCRs, forms part of the calcitonin receptors family that also comprises the amylin (AMY_1, AMY_2, and AMY_3) and adrenomedullin (AM_1 and AM_2) receptors [15,92,93]. These receptors are atypical GPCRs that require the heterodimerization of a GPCR with a single-membrane protein, namely, the receptor activity-modifying protein (RAMP) [15,89,92,93] (see Figure 2). Hence, the CGRP receptor is a protein complex between the calcitonin receptor-like receptor (CLR, the GPCR) and RAMP1. To form the AMY_1 receptor, the conjunction of the calcitonin receptor (CTR, the GPCR) with the RAMP1 is essential. Furthermore, the CLR is necessary to form the AM_1 (CLR + RAMP2) or AM_2 (CLR + RAMP3) receptors [15,92,93]. This final high degree of similitude between receptors (as depicted in Figure 2) explains in part the pleiotropic effects induced by CGRP, amylin, and adrenomedullin. For example, in SK-N-MC cells expressing human CGRP receptors, these ligands induce cAMP increases [91]. Besides, using primary rat cultured trigeminal neurons and COS-7 cells transfected with rat or human CLR/RAMP1 or CTR/RAMP1 proteins, Walker et al. [94] showed that amylin and CGRP stimulate the AMY_1 receptor with equal potency [94].

Moreover, in HEK cells, Bhakta *et al*. [91] showed that erenumab (but not fremanezumab) not only binds to the CGRP receptor and promotes its internalization, but it also antagonizes the amylin-induced cAMP signaling at the AMY_1 receptor and promotes its internalization (a process that can also affect the pharmacokinetics of erenumab). On this basis: (i) it seems that RAMP1 is relevant for the action of erenumab [95]; and (ii) the interaction of erenumab with the AMY_1 receptor underscores the emergence of some potential unexpected side-effects elicited by this mAb. This view gains weight when considering that the amylin system has a relevant function in the hedonic control of eating and that amylin analogues have been designed to treat obesity [96,97]. In contrast, fremanezumab, which directly sequesters CGRP molecules and does not bind to the CGRP receptor: (i) cannot induce internalization of the CGRP receptor; and (ii) only blocks the CGRP receptor activation induced by CGRP, but not by adrenomedullin or intermedin [91].

In summary, the above data suggest that not only do mAbs (and gepants as well) disrupt the CGRPergic function, but they also exert some activity in parallel systems as cross-receptor pharmacology cannot be avoided. This cross-reactivity may be relevant in the case of long-term use of compounds that block the CGRPergic system.

4.2. Safety and efficacy of current mAbs in antimigraine therapy

4.2.1. Advantages and concerns in the use of anti-CGRPergic mAbs

The potential side-effects induced by a long-term blockade of the CGRPergic system have been recently reviewed [11]. Certainly, apart from cross-receptor pharmacology and the production of autoantibodies against the mAbs [98], the main concern is that CGRP exerts some cardioprotective effects [21]. However, as previously pointed out, several advantages seem to overpass some concerns in the use of mAbs. One of the most relevant ones (due to the experience with gepants) is that these molecules are not hepatotoxic [10,22,44,72]. Furthermore, due to their high molecular weight and structure, their therapeutic effects seem to be mainly related to peripheral actions; consequently, CNS side effects can be ruled out. In addition, given the long half-lives of mAbs, it seems more feasible to maintain a greater adherence to the treatment schedule, as one monthly injection reduces migraine attacks and the associated symptoms. Most importantly, these types of molecules do not appear to induce medication overuse headache (MOH), which, on the contrary, can be produced by some specific (*e.g*. ergots, triptans) and nonspecific (*e.g*. NSAIDs) acute antimigraine treatments [9,99]. In contrast, the main concerns about the long-term use of mAbs include the following: (i) a possible interference with cardiovascular homeostasis because of the potential role of CGRP in cardiovascular modulation [21]; (ii) a cross-reactivity with the AMY_1 receptors [91], as discussed above; and (iii) an induction of immunogenicity, a general concern in this type of biological therapy [75].

Regarding the possible interference of CGRPergic mAbs with cardiovascular homeostasis, CGRP is a potent vasodilator with cardioprotective functions and, consequently, the impact on cardiovascular function remains a serious concern [21]. Notwithstanding, relevant cardiovascular effects have not yet been reported. Indeed, in a 5-year open-label extension study, treatment with erenumab appears to be not only effective but also safe [52]. In this sense, some data seem to suggest that CGRP does not play a relevant role in modulating cardiovascular function [100]; nevertheless, under cardiovascular dysfunction, the CGRPergic system seems to play a critical role as a protective molecule [21]. In keeping with this view, de Boer *et al*. [101] has recently suggested that blockade of the CGRPergic system in patients with small vessel disease increased the probability of a cerebral ischemic event. Besides, erenumab *per se* showed no direct vascular effects, but this mAb inhibits the CGRP-induced vasodilation of human peripheral blood vessels, including the middle meningeal, internal mammary, and coronary arteries) [102].

Finally, as far as immunogenicity is concerned, overall, all therapies involving mAbs can potentially induce the biosynthesis of autoantibodies, a fact that limits the therapeutic utility, efficacy, and safety of mAbs. The autoantibodies induced by these immunogens can be divided into neutralizing (blocking the activity of the mAbs) or non-neutralizing (without effect on the mAb activity) [103]. In the case of mAbs against migraine, the incidence of migraineurs developing antibodies against mAbs is highly variable (<1% – 18%). Indeed, phase 2 or 3 RCT's showed that antibodies against mAbs are elicited in: (i) 16–18% of patients treated with eptinezumab [63,64]; (ii) 3–5% of patients treated with galcanezumab [104,105]; (iii) 4–8% of patients treated with erenumab [106,107]; and (iv) 0.3–0.7% in patients treated with fremanezumab [60,108].

4.2.2. A general overview of the randomized clinical trials (RCT's) with mAbs against the CGRPergic system

In general, all RCT's have shown that current antimigraine mAbs are safe and that the main adverse event is related to pain at the injection site [12]. Furthermore, if we consider some phase 3 RCT's, all have shown that erenumab, fremanezumab, galcanezumab, and eptinezumab are statistically effective against migraine headache. Briefly, these mAbs decrease the MMD in patients diagnosed with episodic [105–109] or chronic [60,64,110–112] migraine. All together, these RCT's showed that the mAbs produced a rapid onset of effect with few adverse events. Certainly, erenumab, fremanezumab, and galcanezumab are effective in patients where other antimigraine treatments have failed [113].

Furthermore, a recent post hoc analysis from three RCT's (trials registration: NCT02614183, NCT02614196, and NCT02614261) consistently showed that galcanezumab reduced the MMD in patients with episodic and chronic migraine. Most specifically, the MMD with prodromal symptoms (excluding aura), nausea and/or vomiting, photophobia, and phonophobia were reduced in both types of patients. In contrast, in patients with chronic migraine with aura, the reduction in MMD was not statistically different. Nevertheless, these results should be viewed with caution as there are some biases related to the method used to collect the data, *i.e*. patients that self-reported the presence or absence of aura [114]. Additionally, when using the same data from the above RCT's, galcanezumab seems to induce an early onset of action, a sustained reduction in MMD, and a gradual loss of effect after cessation of treatment [115].

On the other hand, one of the main striking observations is the high placebo response observed in RCT's; indeed, the median persisting MMD in patients treated with mAbs was 55%, while in placebo, it was 75% [116]. Although we could think that the net effect in reducing MMD is minimal, there exists a lack of well-established comparisons between the therapeutic effects of mAbs versus established prophylactic treatments against migraine [116,117]. In this sense, a recent study considering the number of patients needed to be treated for a patient to achieve ≥ a 50% reduction in migraine days, the analysis showed that mAbs exhibited a slightly better benefit-risk ratio than those of the established treatments (*i.e.* topiramate, propranolol, and onabotulinum toxin A) [117].

4.2.3. Current data about the mAbs against PACAPergic transmission in migraine therapy

Since only ~25% of migraine patients treated with CGRPergic mAbs have responded to these novel therapeutic strategies, the potential role of other neuromediators is suggested. In keeping with this suggestion, some studies indicate that the pituitary adenylate cyclase-activating peptide (PACAP) plays a role in migraine [118–120], and two mAbs against the PACAPergic transmission have been developed. Indeed, AMG 301 (zelminemab), a mAb against the PAC_1 receptor, was recently tested; unfortunately, the phase 2 RCT showed that zelminemab provided no beneficial preventive effects over placebo [121]. It is essential to highlight that the utility of PAC_1 neutralizing tools against headaches and migraine is a novel approach that requires further understanding at the preclinical level [122]. On the other hand, a mAb against PACAP-38 (a potent vasodilator of cranial blood vessels), namely, ALD1910 [123], was developed in an attempt to provide an alternative treatment for those whose anti-CGRPergic medications resulted ineffective [124]. In the latter case, since infusions of PACAP-38 may induce headache, vasodilatation, and tachycardia in healthy subjects and patients with migraine [118,125], its blockade seems promissory. However, the exact mechanisms by which PACAP-38 produces migraine attacks remain to be characterized [126].

Ultimately, migraine involves a complex pathophysiology in which, apart from the role of CGRP, several other neuromodulators also play a role [12,127,128]. Despite all the difficulties and challenges that mAbs technology brings, they are powerful, highly selective, and prospective medications that should increase our therapeutic arsenal against various diseases, including migraine.

5. Expert opinion

Migraine is a common and chronic neurovascular disorder. Although abortive pharmacotherapy is indicated to alleviate this type of pain, when migraine attacks occur four or more times in a month, the use of preventative (prophylactic) therapy is encouraged [129,130]. In this sense, the current first-line of prophylactic medications against migraine are nonspecific and included, among others, β-blockers, Ca^{2+} channel blockers, and antiepileptic drugs [9]. Certainly, before the advent of mAbs against the CGRPergic transmission as specific prophylactic drugs, Sandoz had previously developed two 'specific' prophylactic medications from 1960 to 1970, presumably working mainly on the 5-HT system [for references see [131,132,133,134], namely: (i) methysergide (currently known as a nonselective antagonist for serotonin $5\text{-HT}_{2A/2B/2C}$ and 5-HT_7 receptors; and (ii) pizotifen (presently recognized as a nonselective antagonist for $5\text{-HT}_{2A/2B/2C}$, $\alpha_{1/2}$-adrenergic, histamine H_1, and muscarinic $M_{1/2/3}$ receptors). Certainly, methysergide and pizotifen had been extensively used as specific prophylactic antimigraine drugs; unfortunately, they fell into disuse due to some concerns about their safety and the availability of other preventives [131–134].

Still, the advent of mAbs against the CGRPergic transmission (a backbone in the pathophysiology of migraine) gives us a new air to attack this disabling disorder. Indeed, we must keep in mind that the search for new molecules lies in the fact that standard pharmacotherapy is not effective in all cases; also, adverse events may play a role [9]. Furthermore, specific acute antimigraine treatments (*e.g.* ergots, triptans, and probably ditans) cannot be chronically used as they have the potential to induce MOH. As previously proposed [134–136], this secondary headache: (i) seems to have its etiology in a dysfunction of pain processing at the level of the CNS; (ii) follows overuse of these treatments; and (iii) may probably be induced by activation of central $5\text{-HT}_{1B/1D/1F}$ receptors. In this context, anti-CGRPergic drugs (gepants and mAbs) seem not to induce MOH. As previously pointed out, the therapeutic effects of these new molecules seem to be restricted to the periphery [9,10,83,84,87,112,137].

Regarding the apparent lack of CNS penetration of the mAbs [87], it is noteworthy that these novel agents are effective in treating migraine with aura, where visual, sensory, and other CNS symptoms are common. Although there is no clear-cut explanation on how mAbs prevent migraine with aura, Schain et al. [138] have demonstrated that fremanezumab: (i) failed to affect dural arterial dilatation and plasma protein extravasation (PPE) induced by cortical spreading depression (CSD); and (ii) blocked dural arterial vasodilatation induced by exogenous CGRP. Together, these data suggest that CSD-induced dural arterial vasodilatation is not mediated by CGRP.

Since, in addition, CSD occurrence remained unaffected by fremanezumab after BBB disruption [139], the antimigraine action of these mAbs seems more likely to be: (i) unrelated to central effects; and (ii) restricted to the periphery (for refs. see [10]). In agreement with these views, current evidence suggests that migraine starts in the CNS (probably at the hypothalamic level), inducing not only a CSD but also a malfunction of the trigeminovascular system. This can result in an overactivity of primary afferent fibers, which, in turn, would release CGRP at the vascular level favoring cranial vasodilation [128,140–142]. This vasodilation stimulates the primary afferent fibers inducing sensitization of the trigeminovascular system and other diencephalic structures. Under this reasoning, inhibiting the CGRPergic transmission at the peripheral level diminishes the probability that the trigeminovascular system will be engaged in a state that a migraine attack occurs. Certainly, outside the CNS, in the trigeminovascular system, the CGRP receptor is preferentially found in Aδ-fibers , whereas the peptide (*i.e.* CGRP) is localized in C-fibers [143]. These data agree with electrophysiological findings showing that fremanezumab inhibits the activity of meningeal Aδ – but not C-fibers [144]. Apart from the effect on Aδ-fibers, this mAb also blocks the CGRP-induced dilation of cerebral and middle meningeal arteries [145].

Beyond the discussion about the relevance of CSD in migraine pathophysiology, we must keep in mind that this neurovascular disorder is complex. Although CGRP seems to play a role, other molecules are also involved [12]. For this reason, we could suggest that therapeutic agents specifically targeting the CGRPergic system are less effective than the triptans because the latter drugs target directly and indirectly several neurotransmission systems resulting (among other actions) in: direct vasoconstriction, a decrease in CGRP release, inhibition of other pronociceptive/vasodilator systems at peripheral and central levels, etc.

Moreover, considering that CGRP may play a role in cardiovascular modulation, the impact of chronic blockade with CGRPergic mAbs highlights the relevance of post-marketing surveillance in mAbs use. Certainly, balancing the pros and cons mainly depends on the health status of the patient, given that cardiovascular diseases [146] and other comorbidities may limit their use. Additionally, in the last year, several publications have consistently reported that CGRPergic mAbs can induce some minor/rare adverse events. For example: (i) CGRP plays a physiological role in the digestive system that explains why anti-CGRPergic pharmacotherapy (*i.e.* erenumab) can induce constipation [147]; (ii) since CGRP triggers the secretion of interleukin-23 by dendritic cells (acting as a key mediator of skin immunity), mAbs may affect protective cutaneous immunity [148]. In the later case, a case report suggest that mAbs (erenumab and galcanezumab) could favor oral candidiasis probably by disruption of the CGRPergic signaling [149].

Finally, it is important to underline that the potential side effects produced by CGRPergic mAbs appear to be, so far, rather minor, as pointed out above. Moreover, it is noteworthy that these novel prophylactic antimigraine agents are not effective in all patients. This suggests that mediators other than CGRP are involved in migraine pathophysiology, as previously discussed [12]; as such, it appears there will be no magic bullet that will guarantee antimigraine effectiveness in all patients.

Funding

This paper was supported by the Consejo Nacional de Ciencia y Tecnología (219707 to C.M.V., and A1-S-23631 to A.G-H.) and the SEP-Cinvestav Research Support Fund (Grant No. 50 to C.M.V.).

Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

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